Containerized Real-Time Stock Prediction System with Drift Detection and Automated Retraining

Group Members

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Problem Statement

Machine learning models deployed in financial markets quickly degrade due to changing market conditions. These changes appear as:

- **Data drift**, where the input feature distribution shifts over time
- **Concept drift** where relationships between inputs and predictions change

This project aims to build a robust, end-to-end machine learning pipeline that not only predicts short-term stock price movements but also actively monitors the health of the model in production.

By integrating a dedicated data drift detection mechanism, the system will automatically flag when shifting market dynamics outdate the model's training data, ensuring long-term reliability and adaptability.

Each Member's Planned Contribution

Dhanvin S : Cloud & Infrastructure Engineer

- **AWS Environment & Security:** Provision the foundational AWS infrastructure (VPC, Security Groups, IAM roles)
- **Containerization:** Dockerize the inference engine and training services, ensuring they are optimized for cloud deployment.
- **Deployment & Scaling:** Deploy the model via **SageMaker real-time endpoints** and perform **Locust load testing** to conduct scalability experiments.
- **Automation:** Implement **AWS Lambda** functions to trigger automated retraining or system alerts based on infrastructure health.

Sarthak Garg : Machine Learning Lead

- **Feature Engineering:** Build the preprocessing pipeline for time-series data, calculating technical indicators like RSI, Moving Averages, and Volume trends.
- **Model Development:** Design, train, and hyperparameter-tune **XGBoost** or **LSTM** models specifically for stock price forecasting.
- **Baseline Definition:** Establish the SageMaker Model Monitor baseline by capturing the statistical "signature" of the training data.
- **Quantitative Analysis:** Define and track ML-specific evaluation metrics (like RMSE or MAE) to benchmark model performance against historical data.

Basil Varghese : MLOps & Systems Monitoring Lead

- **Real-time Monitoring:** Configure **SageMaker Data Capture** and **Model Monitor** to track live inference data.
- **Drift Detection:** Develop custom scripts using statistical tests (e.g., Kolmogorov-Smirnov) to identify feature degradation and data drift.
- **Observability Dashboard:** Build a **Streamlit/Dash** frontend that visualizes live predictions vs. actual prices alongside real-time drift metrics.
- **System Metrics:** Configure **CloudWatch Alarms** for both system health (CPU/Memory) and ML health (Drift/Accuracy), collecting data for the final performance report.

High-Level Architecture

1. **Infrastructure & Deployment:** The core prediction API and monitoring services will be packaged within Docker containers and deployed onto AWS EC2 instances. This ensures a consistent, isolated environment from development to production.
2. **Data Ingestion:** Real-time and historical financial data will be pulled via APIs (such as Alpaca, Yahoo Finance, or Alpha Vantage).
3. **Modeling:** Time-series forecasting models (such as LSTMs or XGBoost) will be used to predict closing prices or intraday movements.
4. **MLOps & Monitoring:** A continuous monitoring service will run alongside the prediction model and track data drift using statistical tests (KS statistic / PSI) to monitor feature degradation.

Novelty

Unlike typical machine learning projects, our work focuses on **deployment, automation, and quantitative infrastructure evaluation rather than only predictive accuracy**.

Our novelty lies in:

- Fully containerized ML deployment in the cloud
- Automated drift-aware retraining loop
- System-level quantitative evaluation of MLOps benefits
- Demonstration of scalability and reliability under real-time streaming load

The project emphasizes the operational value of containerized ML systems.

Implementation Tools

Cloud Services

- Yahoo Finance, or Alpha Vantage API
- Amazon SageMaker (Training, Endpoint, Model Monitor)
- Amazon S3
- Amazon CloudWatch
- AWS Lambda
- Amazon EC2

Containerization & DevOps

- Docker
- GitHub Actions (for CI/CD)

Machine Learning Stack

- XGBoost or PyTorch
- Scikit-learn
- Pandas / NumPy

Evaluation Tools

- Locust (load testing)
- CloudWatch metrics
- Statistical drift metrics (KS test, PSI)